

Setting up remote Linux workstation for IIC-OSIC-TOOLS

Ubuntu 24.04.4 LTS (Noble Numbat) setup

1. Download Desktop Image; we got one from [here](#)
2. Create a bootable USB stick with that image (using preferred software; we used win32_disc_imager on Windows or dd on linux)
3. Restart PC and boot the Ubuntu from USB stick
4. Get through installation process with preferred software
 - we used default version
 - with 2 divided partitions (SATA Disc 240GB):
 - one for Ubuntu (100GB; /)
 - second one for software (~140GB; /home)
 - there is third one default for boot (around 1GB; /boot/uefi)
5. Partitions can be divided after the setup with the help of USB stick by booting Ubuntu from USB and dividing partitions through "Discs" manager

Remote connection to server

Here we mostly followed this [guide](#)

1. Its good to update system packages first:

```
sudo apt update && sudo apt upgrade -y
```

1. Install Vim, OpenSSH, Xrdp and XFCE. Enable ssh and change config - In 'sshd_config' file uncomment line that has `#Port 22` in it and change the Port number to designated one. In our case it was ports 10160-10164.

```
sudo apt install vim
sudo apt install openssh-server -y
sudo apt install xrdp
sudo apt install xfce4 xfce4-goodies -y

sudo systemctl enable ssh
sudo ufw allow ssh

cd /etc/ssh
sudo vim sshd_config
# inside the file change uncomment Port line and change the number to your
port
```

3. Reload system services and apply SSH configuration:

```
usystemctl daemon-reload
systemctl restart ssh.socket
```

4. Configure XRDP to use XFCE:

```
sudo adduser xrdp ssl-cert
echo "xfce-session" | tee ~/.xsession
sudo systemctl restart xrdp
```

IMPORTANT: If you have changed your default sshd port, you must update your firewall rules to allow traffic through the new port.

```
sudo ufw allow [port_number]/tcp # change the [port_number] to your port
sudo ufw deny 22/tcp
```

Update firewall rules to enable RDP access:

```
sudo ufw allow 3389/tcp
sudo ufw enable
sudo ufw reload
```

5. In new terminal you can check if everything is working correctly; **IMPORTANT:** remember to logout after each login.

- '-p [port_number]' is port number - change to your own
- '-l [login]' is login - change to your own

```
ssh -p 22 localhost
ssh 192.168.0.160 -p [port_number]
ssh 149.156.107.197 -p [port_number] -l [login]

ssh -L 3389:localhost:3389 [login]@149.156.107.197 -p [port_number]
```

6. If everything is working fine from the side of the Ubuntu now you can connect from the other device.

7. First go into **Powershell** and establish the secure SSH tunnel:

```
ssh -L 3389:localhost:3389 [login]@149.156.107.197 -p [port_number]
```

1. Open **Remote Desktop Connection** and configure settings:

- **Default/Computer:** 127.0.0.1
- **Default/Username:** [login]
- **Display/Display Configuration:** set to your preferred display resolution
- **Display/Colors:** set to True Color or 16-bit High Color
- **Experience/Performance:** Automatic or your preferred connection quality

2. Click **Connect** and log into your remote desktop.

(Optional) VSCode setup

You can follow this quick [guide](#) or download VSC from App Center.

(Optional) Git quick setup

1. Check for updates and install git.

```
sudo apt update
sudo apt install git -y
```

2. Set your identity.

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

3. Generate new ssh key.

```
ssh-keygen -t ed25519 -C "your.email@example.com"
```

4. Start the SSH agent in the background and add new key.

```
eval "$(ssh-agent -s)"
ssh-add ~/.ssh/id_ed25519
```

5. Add ssh key to your github. Firstly copy the output of:

```
cat ~/.ssh/id_ed25519.pub
```

Create new ssh key on github website; Go into Account Settings -> SSH and GPG keys -> New SSH key. Then you can paste the **cat** output you previously copied. Everything should be set now.

6. Now you can test the connection.

```
ssh -T git@github.com
```

If it asks if you want to continue connecting, type **yes** and hit Enter. You should see a success message containing your username. Congrats, everything is working correctly.